ARUN MEHRA

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Professional Summary

Backend Software Developer with 9+ years of experience specializing in Python, Go, and Cloud Computing (AWS, GCP, Azure). Proven track record in building scalable microservices, Generative AI integrations (LangChain, OpenAI API), and IoT platforms with security-first architecture. Expertise in DevOps automation (Docker, Kubernetes, Terraform) and Identity & Access Management (OAuth2, JWT, RBAC). Adept at applying Test-Driven Development, AI-driven automation, and system optimization to enterprise-scale solutions.

Core Skills & Technologies

- **Programming**: Python, Golang, Linux Scripting, Rust, Nodejs, React
- **Generative AI & ML:** LangChain, OpenAI API, Google Vertex AI, LLM Integration, AI-driven Automation
- **Databases**: MySQL, PostgreSQL, MongoDB, Cassandra, SQLite
- **Frameworks & Libraries**: Flask, Django, FastAPI, Pytest, Streamlit, Scrapy, Pandas, NumPy, OpenCV
- **Cloud Platforms**: AWS, GCP, Azure
- **DevOps & CI/CD**: Docker, Kubernetes, Jenkins, Terraform, Prometheus, Grafana, Kafka, ELK Stack
- **Security & IAM**: OAuth2, OpenID, JWT, RBAC
- **Protocols**: REST, SOAP, MQTT, HTTP, IoT Matter Protocol
- **Monitoring & Analytics**: AWS Kinesis, OpenTelemetry, Prometheus, New Relic, QuickSight
- **Version Control & Agile Tools**: Git, Bitbucket, Jira, Confluence, Scrum

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Professional Experience

Lead Software Developer

Panasonic India Pvt. Ltd. | Oct 2021 – Dec 2025

Technologies: Python, Go, Flask, FastAPI, LangChain, OpenAI API, Google Cloud Vertex AI, Redis, Kafka, AWS Lambda, IoT Core, Mongo DB, Kafka, Kinesis, Google Cloud Run

- Spearheaded the integration of Generative AI (LLM-based assistants) into Panasonic’s IoT ecosystem, enabling intelligent automation, natural language support, and personalized customer experiences.
- Integrated LangChain with IoT device APIs to provide contextual recommendations (energy saving, predictive maintenance, usage analytics).

- Led the development of MirAle, a cloud-based IoT platform integrating the Matter Protocol for cross-brand device interoperability.
- Designed and deployed scalable microservices architecture for device and home management, ensuring high availability and reliability.
- Developed IAM solutions with Role-Based Access Control (RBAC) to enhance security and multi-tenant access management.
- Implemented FOTA/LOTA updates via AWS Lambda and MQTT, streamlining remote firmware upgrades and reducing operational overhead.
- Led DevOps & Security teams to optimize AWS cloud security policies, improving incident response efficiency and system reliability.
- Deployed containerized microservices on Google Cloud Run, ensuring cost-efficient scalability and streamlined production workloads.

Software Engineer

FireEye Cyber Security | Oct 2020 – Oct 2021

Technologies: Python, Go, Celery, Django, Redis, PostgreSQL, AWS (Lambda, S3, Kinesis, CloudWatch, QuickSight), New Relic, Threat Analytics

- Developed Cloud SIEM reporting tools using GraphQL & MongoDB, automating security incident detection and response.
- Designed and implemented an AWS Cloud cost optimization tool, leveraging machine learning-based anomaly detection to optimize resource usage.
- Led collaboration with DevOps & Security teams, implementing SIEM monitoring policies to align with compliance standards and enhance security posture.

Software Developer

Sierra Circuits | Jun 2019 – Sep 2020

Technologies: Python, KiCad, WxPython, Flask, FastAPI, Cassandra, AWS Lambda, AWS API Gateway, AWS S3

- Developed a KiCad plugin that automated PCB design validation, reducing design errors and improving accuracy.
- Designed and deployed Python-based microservices for PCB automation, streamlining manufacturing workflows and reducing delays.
- Created Flask-based PCB analysis tools, accelerating design verification and validation processes.

Senior Technology Consultant

Aequm India Pvt. Ltd. | Jun 2016 – Jun 2019

Technologies: Python, Flask, FastAPI, MySQL, SQLite, Pandas, NumPy, OpenCV, Tesseract OCR, SOAP APIs

- Developed Retail POS microservices, enabling real-time data synchronization across multiple store locations.
- Designed AI-based invoice extraction models using OpenCV and TensorFlow to automate invoice processing.
- Implemented CI/CD pipelines with Jenkins and AWS CodePipeline, optimizing deployment workflows and reducing manual intervention.

Key Projects

MirAle

Technologies: Python, Go, Flask, FastAPI, Redis, MQTT, Kafka, Kinesis, AWS (Lambda, EC2, CloudWatch, IoT Core), Google Cloud, React, LangChain, OpenAI API, Vertex AI, Google Cloud Run

- Designed and implemented a cloud-native, microservices-based backend architecture to support large-scale IoT device onboarding, lifecycle management, and real-time communication across heterogeneous devices using the Matter Protocol.
- Architected loosely coupled backend microservices in Python and Go using Flask and FastAPI, covering device management, user & tenant management, IAM, telemetry ingestion, analytics, and automation services.
- Designed an API-first system architecture, exposing secure RESTful APIs over HTTP with clear service boundaries, versioning, and backward compatibility to support web (React), mobile, and third-party integrations.
- Implemented an event-driven system design using Kafka and AWS Kinesis to process high-throughput IoT telemetry streams, enabling asynchronous workflows, near real-time analytics, and system decoupling.
- Built a Virtual Device Service capable of simulating thousands of devices over MQTT, enabling load testing, fault injection, scalability validation, and pre-production system verification.
- Designed a multi-tenant backend architecture with OAuth2, JWT, and RBAC, ensuring secure and isolated access for users, devices, and applications across tenants.
- Implemented scalable data persistence strategies, using MongoDB for device metadata and configurations, PostgreSQL/MySQL for transactional and relational data, Redis for caching, session handling, and low-latency access.
- Designed and implemented asynchronous firmware update pipelines (FOTA/LOTA) using AWS Lambda and MQTT, ensuring reliable remote updates with minimal device downtime.
- Integrated Generative AI capabilities using LangChain, OpenAI API, and Google Vertex AI to power intelligent assistants for natural language device interaction, contextual recommendations, energy optimization, and predictive maintenance insights.
- Containerized backend services using Docker and deployed them on Google Cloud Run and Kubernetes-based environments, enabling autoscaling, rolling deployments, and cost-efficient production workloads.
- Implemented backend observability and resilience patterns, leveraging AWS CloudWatch, structured logging, metrics, retries, and graceful degradation to ensure system stability in distributed environments.
- Collaborated with frontend, DevOps, security, and product teams to define end-to-end system design, API contracts, deployment strategies, and operational workflows for enterprise-scale IoT solutions.

Helix XDR (Cloud SIEM Solution)

Technologies: Python, Go, Celery, Django, Redis, PostgreSQL, AWS (Lambda, S3, Kinesis, CloudWatch, QuickSight), New Relic, Threat Analytics

- Developed secure APIs for UEBA (User and Entity Behavior Analytics) tools, improving threat detection capabilities and automated incident response.
- Built real-time monitoring dashboards using New Relic and AWS QuickSight, enhancing system observability, performance monitoring, and security insights.
- Designed and implemented Threat Analytics pipelines leveraging machine learning models to detect anomalous activities, intrusion attempts, and insider threats.
- Integrated AWS Kinesis and CloudWatch for real-time log streaming and security event processing, improving incident detection speed and response times.
- Optimized data ingestion and correlation engines, reducing false positives in threat detection and improving SIEM analytics efficiency.
- Collaborated with security and DevOps teams to refine SIEM monitoring policies, ensuring compliance with industry standards and security best practices.

Sierra Companion App

Technologies: Python, KiCad, WxPython, Flask, FastAPI, Cassandra, AWS Lambda, AWS API Gateway, AWS S3

- Developed a real-time PCB design validation plugin using KiCad and WxPython, improving design accuracy and reducing human errors.
- Designed and implemented microservices-based architecture for PCB design automation, ensuring scalability and modularity.
- Built serverless APIs using AWS Lambda & API Gateway, enabling on-demand processing while reducing infrastructure costs.
- Integrated PCB design libraries with Python-based automation scripts, streamlining design verification and validation processes.
- Utilized AWS S3 and Cassandra for storing and retrieving design schematics, ensuring efficient data access and version control.

99POS System (Retail POS Solution)

Technologies: Python, Flask, FastAPI, MySQL, SQLite, Pandas, NumPy, OpenCV, Tesseract OCR, SOAP APIs

- Developed a comprehensive Point of Sale (POS) system with billing, inventory, and sales management features, improving retail operations and transaction efficiency.
- Designed and implemented RESTful APIs using Flask and FastAPI to facilitate seamless interaction between the POS system, inventory database, and external services.
- Integrated OCR-based document scanning using OpenCV & Tesseract, automating invoice and receipt processing, reducing manual data entry errors.

- Built a robust inventory tracking system with real-time stock updates, order processing, and sales forecasting using MySQL and SQLite.

Education

- PG Diploma in Embedded System Design – CDAC Bangalore (Aug 2015)
- B.E. in Electronics & Communication Engineering – Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal (June 2014)